Improving Retrieval of Sea Surface Roughness From Landsat-8 OLI and Sentinel-2 MSI Imagery by Eliminating for Ambiguity Effects

Chen Wang, Huaguo Zhang, Guanghong Liao, Wenting Cao, Juan Wang, Dongling Li, and Xiulin Lou

***Abstract*— Sea surface roughness (SSR) is a common metric used to characterize the condition of sea surfaces. Sun glitter (SG) is the reflected sunlight off the fluctuating sea surface, with its distribution and intensity markedly influenced by SSR. However, the estimation of SSR through single-angle SG images may produce ambiguity, due to the assumptions applied in the underlying statistical model describing the orientation of the sea surface facets as a function of wind speed. In this study, we investigate the causes of SSR ambiguity and its distribution characteristics via simulation experiments. Based on these findings, we propose a method for identifying critical points of ambiguity and eliminating them for SSR estimation, leveraging single-angle SG images. This approach enables pixel-scale adaptive quantitative inversion of SSR. To validate our model, publicly available global high-resolution SG images (near-infrared (NIR) and shortwave infrared (SWIR) bands) captured by the Landsat-8 Operational Land Imager (OLI) and Sentinel-2 MultiSpectral Instrument sensors are utilized for SSR estimation. Subsequently, the modal values of SSR are expressed as wind-speed values for accuracy assessment. The estimated wind speeds exhibit good agreement with reanalysis wind data from the European Centre for Medium-Range Weather Forecasting (ECMWF) and buoy wind data from the National Data Buoy Center (NDBC). Meanwhile, the comparison results of the two bands are also very consistent, indicating that the method is stable for application on NIR and SWIR bands. The model's efficacy is demonstrated using typical oceanographic scenarios from two regions: the Timor Sea and the English Channel. The resulting SSR images depict detailed oceanographic features with minimal noise, showcasing quantitative SSR changes at the pixel level. These outcomes underscore the effectiveness of the proposed method in resolving SSR ambiguity inherent in widely used optical remote sensing images such as Landsat-8 and Sentinel-2, thereby enhancing SSR retrieval accuracy.**

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I. Introduction

SEA-SURFACE roughness (SSR), which characterizes microscale and small-scale sea surface undulations (or roughness) [1], is pivotal in characterizing ocean dynamics and examining air–sea interactions [2], [3]. However, directly measuring SSR poses challenges, owing to the complexity and randomness of sea surface undulations. In response to this challenge, remote sensing approaches modulated by SSR have been developed, which encompass synthetic aperture radar (SAR) [4], [5], [6], [7] and Sun glitter (SG) [1], [8], [9], [10], [11].

In SAR imagery, the backscatter coefficient is utilized to represent the distribution and intensity of the SSR [12], [13]. Brighter reflected signals indicate greater SSR, whereas relatively darker signals indicate low sea surface roughness (or a smooth sea surface). Hwang et al. [14] and Hwang and Fois [15] proposed a method to invert SSR for different wind conditions (low and high wind speeds) using radar backscattering coefficients of Ku-, C-, and L-band. The results of their study can be used for quantitative evaluation of SSR spectral models and improve the applicable wind-speed range for microwave scattering and emission calculations.

Compared with SAR, the SSR modulation process of SG is relatively complex. Cox and Munk [1] developed a mathematical model (CM model) relating SG radiation to the mean square slope generated by wind (correlated with SSR), employing a wave-slope probability distribution function (pdf) to statistically determine the relationship between SSR and sea surface wind speed. This model laid the foundation for SSR and wind-speed retrieval using SG images. Levanon [9] retrieved SSR and sea surface wind speeds at three Pacific Ocean locations using continuous SG images acquired by geosynchronous satellites, based on the width of large-scale SG area fitting. Wylie et al. [16] explored the accuracy and spatial distribution of sea surface wind-speed inversion based on SG brightness fitting at spatial points. They concluded that the SG method is capable of providing very detailed spatial coverage of wind speed around coastal areas and islands. Fox et al. [17] investigated the relationship between multiangle

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SG patterns obtained by MISR (nine observation angles) and sea surface wind speed, employing the multiangle SG fitting method. They introduced the concept of ambiguity, resolving it by selecting the minimum observed absolute reflectance at nine observation angles. Jackson and Alpers [18] investigated the characterization of sensor zenith angles for different SSRs to produce the same SG radiance and proposed critical sensor viewing angles. Subsequently, a model for estimating high-resolution SSR at the pixel scale based on SG images of two observation angles was proposed by Zhang et al. [11]. They utilized the ratio of SG images of two angles to eliminate ambiguity. To date, retrieving SSR at the pixel scale using single-angle SG images remains limited by ambiguity, posing a significant challenge in remote sensing research.

In this study, we investigate the causes of SSR ambiguity and its distribution characteristics through simulation experiments. Based on our findings, we propose a method for identifying and eliminating critical points of ambiguity in SSR estimation, utilizing single-angle SG images. This method facilitates pixel-scale adaptive quantitative inversion of SSR. Furthermore, we validate the SSR estimates using SG images sourced from publicly available global high-resolution optical satellite imagery (Landsat-8 and Sentinel-2). To demonstrate the model's efficacy, we compare the estimation results with reanalysis and buoy wind data, as well as results from both bands. Subsequently, we provide illustrative examples of high spatial resolution SSR at the pixel scale by examining typical cases in two regions. Finally, we discuss the sensitivity of the seawater refractive index, the uncertainty associated with the SSR ambiguity elimination method, and the influence of observation geometries that lead to possible ambiguities for Landsat-8 and Sentinel-2.

II. Data Sources and Processing

A. Satellite Imagery

Satellite images are obtained from the Operational Land Imager (OLI) on Landsat-8 spanning from 2013 to 2020, and the MultiSpectral Imager (MSI) on both Sentinel-2A (2015–2020) and Sentinel-2B (2017–2020). In this study, SG images are utilized to invert SSR. Therefore, images located at low and middle latitudes are chosen because satellites can easily capture SG signals. In addition, to validate the applicability of the methodology, images are selected from multiple ocean areas in both the northern and southern hemispheres. These satellite images predominantly cover the Gulf of Mexico, the Caribbean Sea, the Mediterranean Sea, the South China Sea, and the Timor Sea, as depicted in Fig. 1(a).

Tavares et al. [19] evaluated the performance of six software programs for atmospheric correction and Sun glitter correction and found that C2X (C2RCC calibrated to turbid waters), ACOLITE, and glint removal for Sentinel-2 (GRS), provided the most consistent surface water reflectance compared to in situ. To be able to automatically correct for atmospheric conditions and extract SG, we chose the ACOLITE software. These images are then processed to SG reflectance (ρ_g) using the ACOLITE software (version 20190324.0). After atmospheric correction processing, reflectance images at

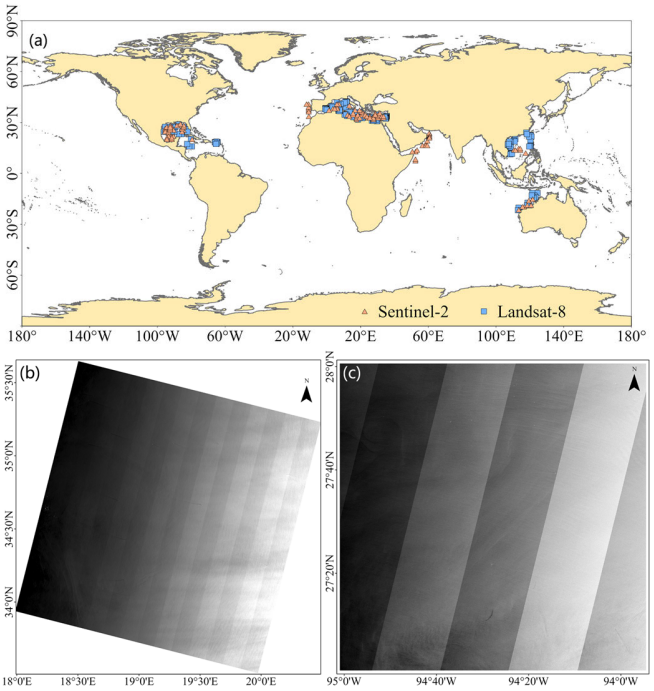


Fig. 1. (a) Distribution of satellite images from Landsat-8 (box) and Sentinel-2 (triangle) used in this study. The high-resolution coastline database is from P. Wessel and W. Smith [23]. (b) ρ_g image in the NIR band. The satellite image acquired by Landsat-8 on June 7, 2016, located in the Mediterranean Sea. (c) ρ_g image in the NIR band. The satellite image acquired by Sentinel-2 on April 9, 2019, located in the Gulf of Mexico. The lack of ground control points resulted in less accurate geo-correction of the images and hence the significant stripes in (b) and (c).

TABLE I
Center Wavelengths (nm) and Spatial Resolution (m) of the Bands Used in This Study for Each Satellite Sensor

Sensor	NIR	SWIR	Spatial resolution
Landsat-8/OLI	865	1609	30
Sentinel-2A/MSI	865	1614	10*
Sentinel-2B/MSI	864	1610	10*

* ACOLITE resampled the Sentinel-2 SWIR band images to 10 m by duplicating the pixels [20].

wavelengths close to the 1600-nm band are used for glint extraction in the other bands since ACOLITE assumes that water-leaving radiance in this band is nearly zero (the method is detailed in [20]). ρ_g in the near-infrared (NIR) band is used for SSR estimation. To verify the consistency of results between bands, ρ_g in the shortwave infrared (SWIR) band is processed simultaneously. The band center wavelengths and spatial resolutions of all the sensors are listed in Table I.

In Fig. 1(b) and (c), we present ρ_g images in the NIR band from the Mediterranean Sea and the Gulf of Mexico, acquired by Landsat-8 on June 7, 2016, and Sentinel-2B on April 9, 2019, respectively. The imaging modules of Landsat-8 and Sentinel-2 use a similar design, with a single image stitched together from several detector modules, 14 for OLI and 12 for MSI [21]. The lack of sea surface control points resulted in low geographic correction accuracy of the images

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and the appearance of distinct stripes in Fig. 1(b) and (c) that corresponded exactly to the imaging module [22].

In this study, the corresponding mask data provided by the U.S. Geological Survey (USGS) and the European Space Agency (ESA) are utilized to mitigate cloud interference in the images. In addition, to address the impact of cloud edges, an additional 10-pixel buffer is applied based on the cloud mask data [24].

B. Other Data

Quantitative validation of SSR is performed by converting SSR to sea surface wind speed based on the linear relationship developed by Cox and Munk [1]. We then select the reanalysis and buoy wind data corresponding to the image.

Reanalysis wind-speed data are sourced from the ERA-5 dataset provided by the European Centre for Medium-Range Weather Forecasting (ECMWF). This dataset offers wind-speed measurements at a height of 10 m above the sea surface, with a spatial resolution of 0.125° × 0.125° and a temporal resolution of 6 h.

The in situ measured wind-speed data corresponding to the imaging are obtained from buoy recorders operated by the National Data Buoy Center (NDBC). Standard meteorological buoy wind data, recorded at 30-min intervals, are utilized. Due to variations in the heights of the anemometers, it is necessary to standardize the buoy wind speeds to a height of 10 m prior to validation analysis. To achieve this, the power-law wind-profile under near-neutral atmospheric stability conditions proposed by Hsu et al. [25] is applied.

III. Model

SSR estimation using SG images from Landsat-8 OLI and Sentinel-2 MSI involves several steps (Fig. 2). The main objective of the modeling is to identify and eliminate ambiguities in SSR.

A. Sun Glitter Theory Based on the Cox–Munk Model

The normalized SG radiance (L_{gn}) is estimated based on the studies of Cox and Munk [1] and Gordon [26] followed by the formulation detailed by Jackson and Alpers [18] and Zhang and Wang [27]. The L_{gn} general formula for an isotropic probability distribution function without wind direction dependence is

$$L_{\mathrm{gn}} = \frac{R(\omega)}{4\cos\theta_v \cos^4\beta} \frac{1}{\pi\left(\sigma_0^2\right)} \exp\left(-\frac{\tan^2\beta}{\sigma_0^2}\right) \tag{1}$$

where θ_v represents the view zenith angle of the sensor, $R(\omega)$ represents the Fresnel reflection coefficient affected by reflection (ω) and refraction (ω') angles from a tilted wave facet, β represents the sea surface tilt angle, and σ_0^2 represents the SSR generated by wind. ω and β can be calculated using θ_v, the solar zenith angle (θ_0), and the relative azimuth ($\Delta\emptyset$)

$$\cos 2\omega = \cos\theta_0 \cos\theta_v + \sin\theta_0 \sin\theta_v \cos\Delta\emptyset \tag{2}$$

$$\tan^2\beta = \frac{\sin^2\theta_0 + \sin^2\theta_v + 2\sin\theta_0 \sin\theta_v \cos\Delta\emptyset}{(\cos\theta_0 + \cos\theta_v)^2}. \tag{3}$$

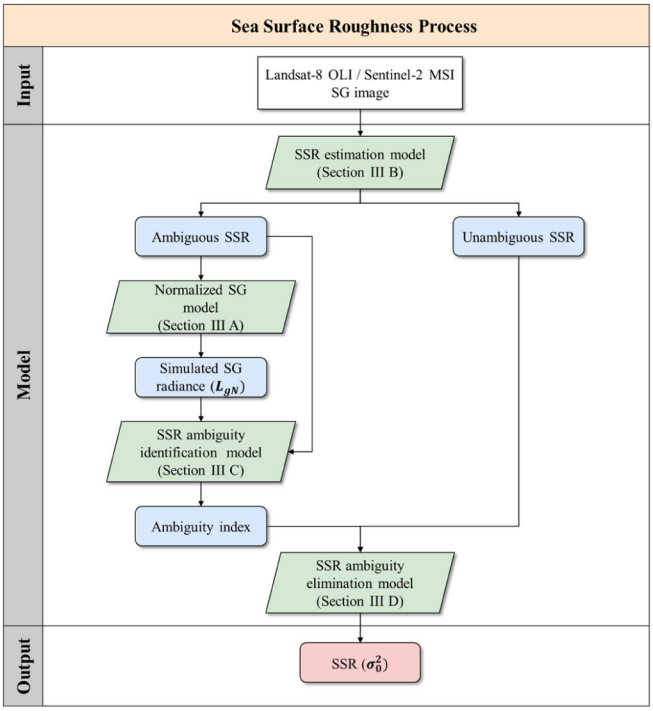


Fig. 2. Methodological steps for adaptive estimation of SSR from SG images.

According to Cox and Munk [1], the relationship between SSR (σ_0^2) and wind speed (W, m/s) at 41 ft (about 12.5 m) above sea level is as follows:

$$\sigma_0^2 = 0.003 + 0.00512W. \tag{4}$$

In this study, Landsat-8 OLI and Sentinel-2 MSI images are processed to ρ_g using ACOLITE. Thus, the conversion of ρ_g and L_{gn} is performed as follows:

$$\rho_g = \frac{\pi L_g}{F_0 \cos\theta_0} \tag{5}$$

$$L_g = F_0 T_0 L_{\mathrm{gn}} \tag{6}$$

where L_g is the Sun glitter radiance, F_0 is the solar irradiance from extraterrestrial, and T_0 is the direct atmosphere transmittance in the Sun-to-water direction.

According to (1)–(3), the geometry angles, SSR, and refractive index are the main parameters that affect the Sun glitter radiance. Based on the above theories and formulas, we can carry out experiments simulating L_{gn}. First, by customizing the parameters (geometry angles, SSR, and refractive index), L_{gn} can be calculated. Second, we can also retrieve SSR based on other parameters (L_{gn}, geometry angles, and refractive index).

B. Estimation Method of Sea Surface Roughness

As detailed in Section III-A, the main parameters affecting SG radiance include geometric angle, SSR, and refractive index of seawater. Among them, the refractive index affects the SG through the Fresnel reflection coefficient. The retrieval of SSR using (1) is nonlinear; therefore, an iterative method is selected. The refractive index of seawater in pristine open-ocean conditions exhibits notable stability across specific

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TABLE II
SIMULATION PARAMETERS

Parameter	Value
sensor zenith angle (θ_v)	4 °
sun zenith angle (θ_0)	23 °
relative azimuth ($\Delta\emptyset$)	200 °
SSR (σ_0^2)	0.003-0.08, step 0.00512

wavelengths [28], [29]. Consequently, we choose the refractive index as a constraint parameter for the iterative process. The SSR of each pixel is estimated using a minimization of the difference between the refractive index and the observed values in the NIR and SWIR bands. We vary the SSR (σ_0^2) in increments of 0.000512 (from 0.003 to 0.08) to determine the difference between the inverted refractive index and the actual refractive index for each pixel. The SSR then corresponds to the minimum of that difference, that is, the SSR of the pixel.

C. Identification of SSR Ambiguity

Ambiguity arises when two distinct SSRs yield identical SG intensities at a Sun-surface-observer observation geometry, rendering SSR retrieval uncertain. Fox et al. [17] considered that ambiguity is an intrinsic characteristic of the reflectance patterns described by the Cox–Munk model, and they addressed it by employing the minimum absolute reflectance observed across nine MISR viewing angles. According to (1), the relationship between SG and SSR is a transcendental equation, and therefore, multiple solutions (no solution, single solution, and double solution) may exist. Zhang et al. [11] eliminated ambiguity by converting the relationship between SG and SSR into an algebraic equation through a multiangle SG ratio. Consequently, we initially identify ambiguity issues via simulation experiments and subsequently propose a method for resolving SSR ambiguity in satellite images with only one observational angle.

We set up a set of parameters (Table II) to analyze the Cox–Munk model and the ambiguity issue of SSR. We divide (1) into three parts as follows:

$$\text{part } 1 = \frac{R(\omega)}{4\cos\theta_v \cos^4\beta} \tag{7}$$

$$\text{part } 2 = \frac{1}{\pi\left(\sigma_0^2\right)} \tag{8}$$

$$\text{part } 3 = \exp\left(-\frac{\tan^2\beta}{\sigma_0^2}\right). \tag{9}$$

Based on the parameters in Table II, we calculate the results for each part as well as for L_{gn}. The value for part 1 is 0.005. The curves of part 2, part 3, and L_{gn} versus SSR are shown in Fig. 3. In Fig. 3(a), we can find that parts 2 and 3 are monotonically decreasing and monotonically increasing functions of SSR, respectively. However, L_{gn} is not a monotonic function of SSR in Fig. 3(b).

In Fig. 3(b), Lgn1 (green dotted line), Lgn2 (red dotted line), and Lgn3 (yellow dotted line) denote Sun glitter of different intensities. The number of intersections of the dotted line with the curve indicates whether ambiguity occurs. When

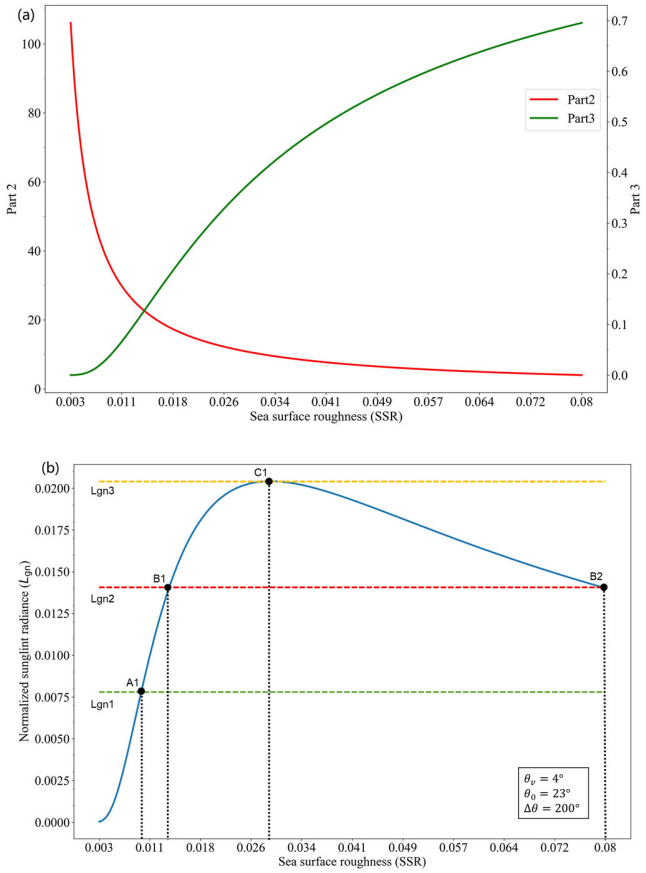


Fig. 3. (a) Curves of parts 2 and 3 versus SSR. (b) L_{gn} versus SSR curve (blue line). Lgn1 (green dotted line), Lgn2 (red dotted line), and Lgn3 (yellow dotted line) denote Sun glitter of different intensities. A–C denote the intersections of the dotted line with the curve.

there is only one intersection, it indicates that there is no ambiguity; when there are two intersections, it indicates that ambiguity occurs at this specific Sun-surface-observer observation geometry. Therefore, when the glitter is less than Lgn2 or equal to Lgn3, no ambiguity occurs; when the glitter is between Lgn2 and Lgn3, ambiguity occurs. For example, Lgn2 has two intersections with the glitter curve, B1 and B2. B1 and B2 correspond to different SSRs. This is the ambiguity phenomenon of SSR. However, which SSR is the true SSR of the pixel is the main focus of this study, i.e., how to identify and remove ambiguities.

The normalized SG radiance (L_{gn}, sr^{-1}) patterns computed for θ_0 of 30° with SSR of 0.029 and 0.054 (corresponding to wind speeds of 5 and 10 m/s) are shown in Fig. 4. The polar angle and polar radius of patterns indicate the relative azimuth ($\Delta\emptyset$, 0°–360°, steps length 1°) and the sensor zenith angle (θ_v, 0°–90°, steps length 1°), respectively. The black dashed line indicates the contour with an L_{gn} intensity of 0.01. The Sun glitter pattern of Fig. 4(b) is broader than that of Fig. 4(a), but the maximum Sun glitter intensity is lower than that of Fig. 4(a). Fig. 4(a) and (b) shows the fact that the Sun glitter is not a simple monotonic function of the sea surface roughness. Jackson and Alpers [18] and Lu et al. [30] analyzed the effect of changes in sensor viewing angle, SSR, and refractive index on the intensity of SG. SSR is then calculated using the method

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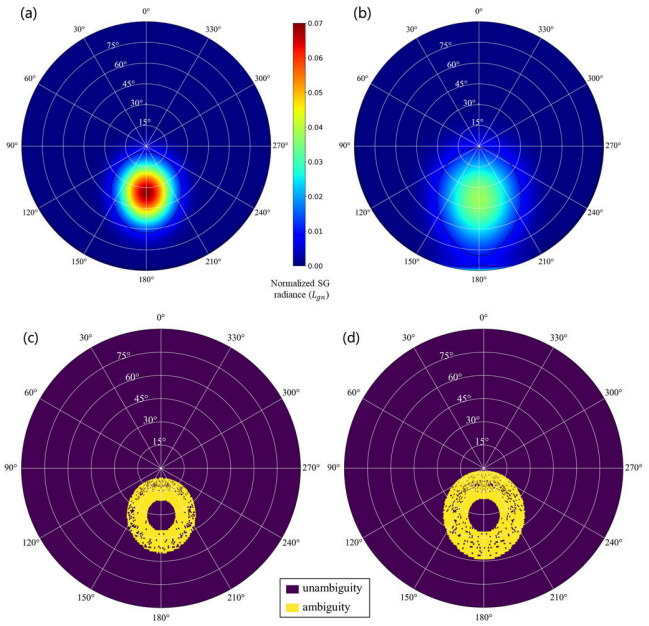


Fig. 4. Normalized SG radiance (L_{gn}) patterns for θ_0 of 30° with an SSR of (a) 0.029 and (b) 0.054, corresponding to surface wind speeds of 5 and 10 m/s, respectively. The polar angle and polar radius of patterns indicate the relative azimuth ($\Delta\emptyset$, 0°–360°, steps length 1°) and the sensor zenith angle (θ_v, 0°–90°, steps length 1°). The black dashed line indicates the contour with an L_{gn} intensity of 0.01. (c) and (d) Ambiguity and unambiguity results of SSR obtained from (a) and (b), respectively. The purple indicates that there is no ambiguity, whereas yellow indicates ambiguity.

described in Section III-B. The number of SSRs retrieved is recorded. If the number is one, it means that there is no ambiguity, and if the number is two, there is ambiguity. The SSR ambiguity results are shown in Fig. 4; purple indicates that there is no ambiguity, whereas yellow indicates ambiguity. We can see that the ranges of the ring-like structures in the polar plots of Fig. 4(c) and (d) are different. It also illustrates that in addition to the observation geometry, the intensity of the glitter is also an important factor in the occurrence of ambiguity. As can be seen, ambiguity is more likely to occur in the broad SG pattern (i.e., high SSR).

To further analyze the ambiguity issue, we selected a Landsat-8 image with which to conduct simulation experiments; the image chosen is the NIR band image from the Gulf of Mexico acquired by Landsat-8 on November 10, 2020 [Fig. 5(a)]. Based on the reanalyzed wind-speed data from the ECMWF, the image corresponds to an average wind speed of 5.15 m/s. Based on the angle data in Fig. 5(a) and an SSR of 0.029, L_{gn} of the simulated image is calculated [Fig. 5(b)]. The SSR is then estimated using L_{gn} based on the method described in Section III-B. Note that the input SSR is 0.029 when simulating glitter, so the retrieved SSR should also be 0.029. However, some areas (pixels) of the image appear ambiguous. Thus, the retrieved SSR appears with two different values, besides 0.029, there is another value. At the ambiguity pixel, two different SSR results (SSR1 and SSR2) were recorded simultaneously. The SSR results are shown in Fig. 5(c) and (d), and each figure is divided into three regions: Region 1 (R1), Region 2 (R2), and Region 3 (R3). The SSR of R1 in Fig. 5(c) and (d) is identical (0.029), indicating no ambiguity. However, R2 (and R3) in Fig. 5(c) and (d) is different, indicating ambiguity, wherein the SSR of R2 in Fig. 5(c) is 0.029, and the SSR of R2 in Fig. 5(d) is greater than 0.029. The SSR of R3 in Fig. 5(c) is less than 0.029, and the SSR of R3 in Fig. 5(d) is 0.029. Based on the input value of SSR (0.029) when simulating the SG, it can be seen that the SSR result should be R1 [Fig. 5(c)/(d)], R2 [Fig. 5(c)], and R3 [Fig. 5(d)].

In order to analyze the profile changes, we export the data along the "Profile" lines in Fig. 5(b)–(d). These data are plotted in Fig. 5(e). The red and green lines in Fig. 5(e) indicate the two results for the inverted SSR. We selected seven positions (p1–p7) in Fig. 5(e) to analyze the relationship between the modeled and derived glitter, as well as the change in the SSR error; the results are shown in Fig. 5(f). On the left side of the point "p2," the red and green lines overlap, indicating that there is no ambiguity, whereas ambiguity is evident on the right side: the low SSR curve (SSR1, red) and the high SSR curve (SSR2, green). For example, the glitter corresponding to position p3 is Lgn3 (value 0.014). C1 and C2 are the two SSRs retrieved using Lgn3; C1 is 0.029 and C2 is 0.057.

In Fig. 5(f), A–G correspond to the SSR results at seven locations, and Lgn1–Lgn7 represent the simulated L_{gn}. The solid lines in Fig. 5(f) indicate the variation curves of L_{gn} versus SSR (0.003–0.08) for the seven positions. From p1 to p7, L_{gn} versus SSR curve changes from short and broad to high and narrow, as shown in Fig. 5(f). Based on the intersection of the solid and dashed lines of the same color in Fig. 5(f), we can clearly understand the phenomenon of ambiguity.

D. Elimination Method of SSR Ambiguity

Based on the SSR input in the simulation experiment, we determine that the actual SSR in the ambiguous area should be B1–C1–D1–E1–F2–G2 in Fig. 5(e) and (f). The SSR of E1 corresponds to position "p5," a pivotal location within the ambiguity area. SSR1 is selected to the left of "p5," and SSR2 is selected to the right. Therefore, we propose a method to resolve the ambiguity of SSR by pinpointing the "p5" positions. The steps were as follows.

Step 1: SSR is estimated at the pixel scale using SG images based on the method described in Section III-B, and two different SSR results are recorded simultaneously for the pixels where ambiguity occurs.

Step 2: The median roughness in the unambiguous region is calculated as the input of the SSR in (1), which is then combined with the geometric angle of the image to calculate the normalized SG irradiance.

Step 3: SSR is estimated using the normalized SG irradiance.

Step 4: The median roughness is then matched with the SSR results from Step 3 to retrieve the critical position "p5." Both SSR1 and SSR2 in the ambiguous area are equal to the median SSR, i.e., the "p5" position.

Step 5: Based on the critical position, the SSR from Step 1 is matched to obtain the final result.

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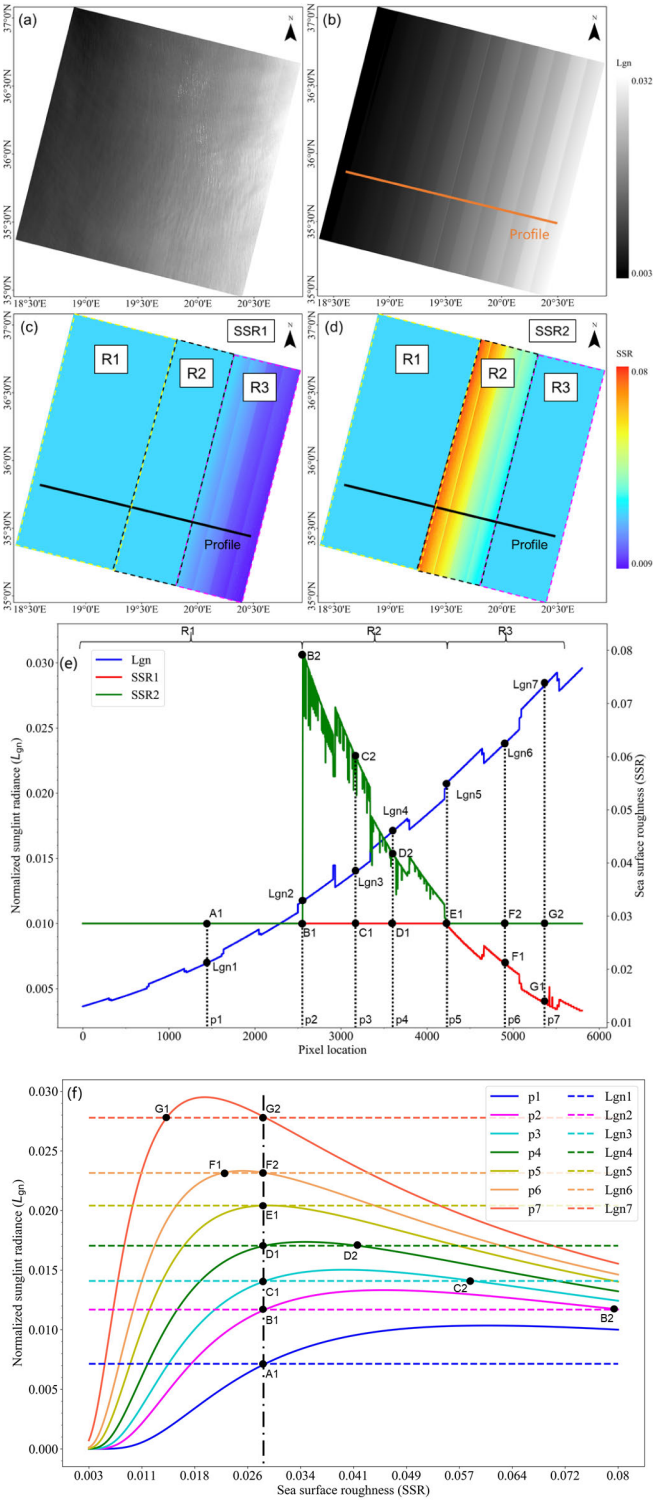


Fig. 5. (a) Landsat-8 NIR band image of the Gulf of Mexico on November 10, 2020. (b) Simulation image of normalized SG radiance. (c) and (d) SSR results with ambiguity. (e) L_{gn} and SSR profile along the line labeled "Profile" in (b)–(d). (f) Relationship between modeled SG and the derived SG at seven positions (p1–p7) in (e), as well as the change of SSR.

Based on the above steps, ambiguity can be eliminated. As is well understood, the actual sea surface is a complex entity influenced by a variety of random fluctuation signals. The spatial distribution of SSR in satellite images appears inconsistent. Consequently, the current method may introduce uncertainty errors in the SSR results, as demonstrated for the actual sea surface case in Section IV-B and discussed in Section V-B. While more accurate methods warrant exploration, they are not the primary focus of this study.

IV. Results and Analysis

A. Estimation Result of Sea Surface Roughness

Following atmospheric correction and cloud masking, the SG images are converted into SSR. The SSR results are categorized into two classes: without ambiguity and with ambiguity. After counting, the number of images without ambiguity is 35, and 256 are ambiguous, accounting for 88%. The results show that it is easy to occur ambiguity in SSR retrieval using Landsat-8 and Sentinel-2 images.

To analyze the ambiguity phenomenon in real SSR images and the effectiveness of the proposed method, we analyze the SSR. The results are shown in Fig. 6. The SG image is derived from Fig. 1(b). Fig. 6(a) and (b) shows SSR1 and SSR2 for ambiguity, respectively, which are similar to those in Fig. 5(c) and (d). The figure is divided into three regions: R1 for unambiguity (yellow box) and R2 and R3 for ambiguity (black and purple boxes). Note that R2 in Fig. 6(a) is different from R2 in Fig. 6(b) (as is R3). In Fig. 6(c), it can be seen the SSR after processing using the method proposed in this study, named SSR3. The SSR results along the "Profile" lines in Fig. 6(a)–(c) are shown in Fig. 6(d), where SSR3 (green line) is filtered with a window of 3 to show the results more clearly. The black lines mark the critical positions extracted in this study, and it can be seen that SSR1 and SSR2 almost overlap; therefore, the processed SSR results are continuous and realistic. Based on the analysis of Fig. 6(d), it is clear that R2 in Fig. 6(b) and R3 in Fig. 6(a) are problematic SSRs. Thus, the performance of the proposed method is determined to be more than satisfactory.

B. Validation Analysis of Sea Surface Roughness

To evaluate the accuracy of SSR, the modal SSR values are converted to wind speeds using (2). Modal SSR values are more suitable for representing background wind-induced SSR [11]. Reanalysis and buoy wind speeds (provided by the ERA-5 dataset and the NDBC, respectively) are used for comparison with the estimated results.

To ensure the accuracy and validity of the wind-speed data, it is first screened and preprocessed. First, the heights at which the estimated and buoy wind speeds were measured are harmonized. The measurement heights for the estimated and reanalysis wind speeds are 41 ft (12.5 m) and 10 m, respectively, whereas that of buoy wind speed depends on anemometer height. Therefore, the estimated and buoy wind speeds are converted to 10 m above the sea surface based on the power-law wind profile proposed by Hsu et al. [25]. Second, the corresponding reanalysis and buoy wind speeds based on latitude, longitude, and imaging time of images are obtained. Modal values of wind speeds in a range consistent with the spatial resolution of the reanalysis wind speeds are selected. For the reanalysis of wind speeds, a resolution of $0.125° \times 0.125°$ corresponds to a range of 400×400 pixels

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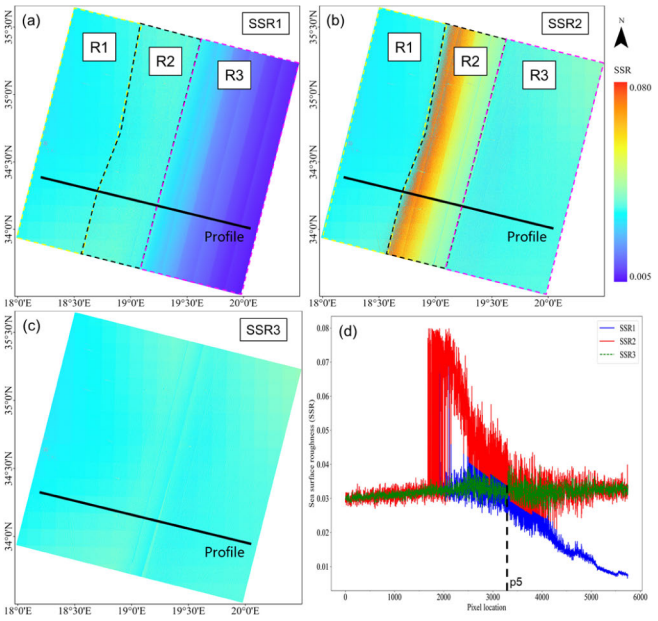


Fig. 6. SSR results derived from SG image of Fig. 1(b). Here, (a) and (b) are SSR results with ambiguity, SSR1 and SSR2, respectively. (c) SSR results after being processed by the method proposed in this study, named SSR3. (d) SSR profile along the "Profile" lines in (a)–(c), where SSR3 (green line) is filtered with a window of 3 to show the results more clearly.

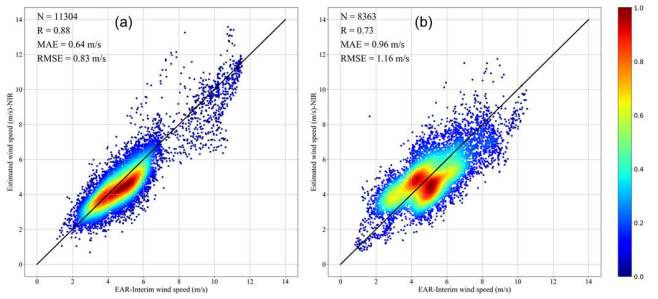


Fig. 7. Scatter density plots of comparison results. (a) Wind speed of Landsat-8 NIR band versus reanalysis wind speed. R is 0.88, MAE is 0.64 m/s, and RMSE is 0.83 m/s. (b) Wind speed of Sentinel-2 NIR band versus reanalysis wind speed. R is 0.73, MAE is 0.96 m/s, and RMSE is 1.16 m/s.

(Landsat-8) or 1200 × 1200 pixels (Sentinel-2). The reanalysis wind speeds corresponding to the imaging time are obtained by interpolation. For the buoy data, the wind speeds record that was within 0.5 h of the imaging time and closest is selected. The modal values of wind speeds within 30 km of the buoy location are selected; this range corresponds to 1000 × 1000 pixels (Landsat-8) or 3000 × 3000 pixels (Sentinel-2). Wind speeds based on the NIR band are compared with the reanalysis and buoy wind speeds in Figs. 7 and 8, respectively. The root-mean-square error (RMSE), correlation coefficient (R), and mean absolute error (MAE) are used to evaluate the estimation results.

As shown in Fig. 7(a) and (b), the wind-speed ranges reanalysis and estimated are 1.14–11.55 m/s and 0.69–13.58 m/s for Landsat-8 NIR band, and 0.83–10.53 m/s and 0.76–11.75 m/s for Sentinel-2. The comparison results show that the estimated and the reanalysis wind speeds matched reasonably well, with R of 0.88 and RMSE of

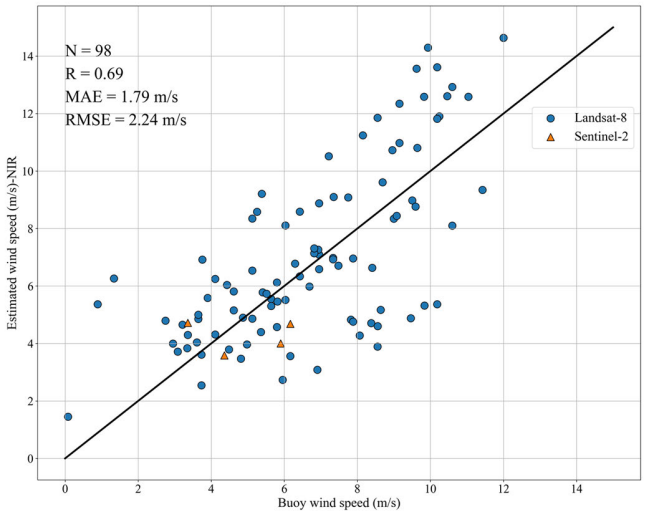


Fig. 8. Wind speed of Landsat-8 and Sentinel-2 NIR band versus buoy wind speed. R is 0.69, MAE is 1.79 m/s, and RMSE is 2.24 m/s.

0.83 m/s for Landsat-8, and R of 0.73 and RMSE of 1.16 m/s for Sentinel-2. The results show that Landsat-8 wind speeds are more accurate than those of Sentinel-2, probably because of the more serious problem of striping in the images of Sentinel-2, which is a result of low-accuracy terrain correction and geographic alignment [31]. The MAEs of the wind speed for Landsat-8 and Sentinel-2 are 0.83 and 1.16 m/s, respectively, which is consistent with the accuracy of the wind speeds estimated by Wylie et al. [16].

As shown in Fig. 8, the range of buoy wind speeds is 0.08–12.0 m/s, and the estimated wind speeds range from 1.45 to 14.64 m/s for Landsat-8 NIR and 3.01 to 4.95 m/s for Sentinel-2 NIR. A comparison of the estimated wind speeds with the buoy wind speeds shows that the RMSE and MAE were 2.24 and 1.79 m/s, respectively. Although the evaluation accuracy of the estimated wind speeds with the buoy wind speeds is lower than that with the reanalysis wind speeds, it is in general agreement with the results of other studies.

The wind-speed results retrieved by Fox et al. [17] using MISR are within 2.4 m/s of the buoy data. The RMSE of 1.0 m/s employing a scatterometer proposed by Vogelzang et al. [33] and an RMSE of 1.119 m/s based on polarized SG images from the PARASOL sensor presented by Harmel and Chami [34], and RMSE of 3 m/s for wind-speed retrieval using SAR images proposed by Yan et al. [31] and Carvajal et al. [32]. Compared to this literature, the inversion accuracy of the method proposed in this study is comparable to that of the scatterometer, but superior to other methods. Differences in data resolution and evaluation methods for each study also lead to some variation in wind-speed inversion accuracy. Based on the above evaluation results, it can be seen that the method proposed in this study can effectively eliminate the ambiguity of SSR and improve the accuracy of high-resolution SSR data.

A comparison of the retrieved wind speed in different bands (NIR and SWIR) is shown in Fig. 9. R of wind speed in the NIR and SWIR bands of Landsat-8 is 0.99, RMSE is 0.28 m/s, and MAE is 0.17 m/s. R of wind speed in the NIR and SWIR

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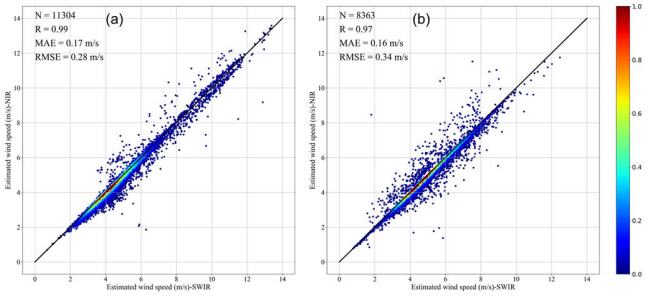


Fig. 9. Scatter density plots of comparison results. (a) Wind-speed results of Landsat-8 NIR versus SWIR. R is 0.99, MAE is 0.17 m/s, and RMSE is 0.28 m/s. (b) Wind-speed results of Sentinel-2 NIR versus SWIR. R is 0.97, MAE is 0.16 m/s, and RMSE is 0.34 m/s.

bands of Sentinel-2 is 0.97, RMSE is 0.34 m/s, and MAE is 0.16 m/s, which indicates that the method is sufficiently stable.

C. Examples of SSR Applications

High-resolution SSR data have been widely used in analyzing sea surface features. Eliminating the ambiguity in SSR improves the precision of SSR data and guarantees the applicability of high-resolution SSR. Typical cases from the two regions are presented as follows.

1) Timor Sea: In Fig. 10, we present an SSR image of the Timor Sea [Fig. 10(b)], acquired by Landsat-8 on December 9, 2013 [Fig. 10(a)], in which the phenomenon of internal waves can be clearly seen. Internal waves cause changes in the sea surface current field, affecting the occurrence of glint, which is observed by satellites [35], [36]. Fig. 10(c) shows a magnified SSR image of the red box in Fig. 10(b), in which it can be clearly seen that the SSR at the internal wave position exhibits bright and dark stripes. Based on the wave pattern, we can determine that the propagation direction of the internal wave is from northwest to southeast.

The SSR of the "Profile" line in Fig. 10(c) can be seen in Fig. 10(d). The average SSR values of the inner wave's front slope, back slope, and background are 0.022, 0.033, and 0.028, respectively [Fig. 10(d)]. On this basis, the internal wave produces an SSR change of 0.011, which accounts for 39% of the wind-generated background SSR. The high-resolution SSR data more clearly reveal the quantitative changes of each part of the internal wave; this information is important for calculating parameters such as the wavelength, amplitude, and depth of the internal wave.

Fig. 11 presents a Sentinel-2A-derived SSR image acquired on October 9, 2020, over the Timor Sea [Fig. 11(a) and (b)], with two coral reefs clearly visible. Based on the distribution of SSR, the SSR values in the inner part of the reef are lower than those in the background. In Fig. 11(c) [a magnified SSR image of the red box in Fig. 11(b)], the SSR on the right side of the coral reef is also lower than the background. Therefore, we can roughly determine that the propagation direction of the waves is from southwest to northeast. In Fig. 11(d) [an SSR profile of the coral reef in Fig. 11(c)], the SSR values at the wave-ward side of the coral reef, within the reef, and at the wave-leaving side of the reef, are approximately 0.033, 0.026, and 0.027, respectively. This suggests that the

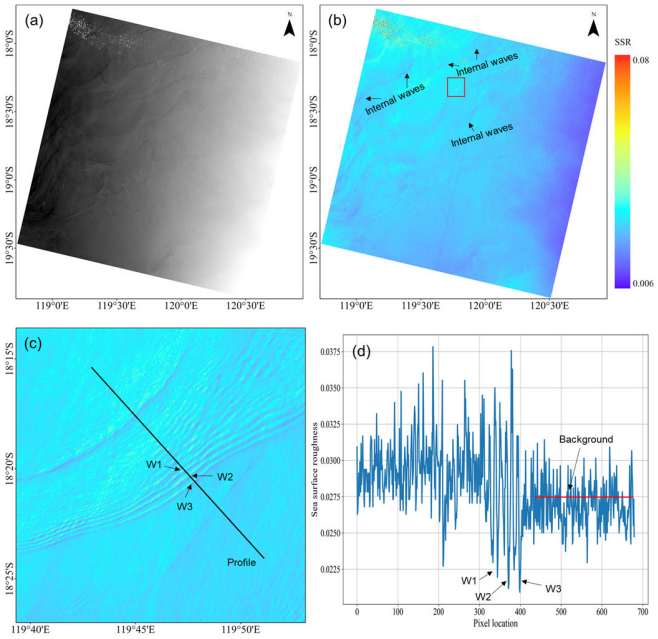


Fig. 10. (a) ρ_g image in the NIR band. The satellite image acquired by Landsat-8 on December 9, 2013, located in the Timor Sea. (b) SSR-estimated image of internal waves. (c) Magnified SSR image of the red box in (b). (d) SSR of the "Profile" line in (c). The red line indicates the average SSR (0.028) of the background region, and labels W1–W3 correspond to the positions of the internal waves in (c).

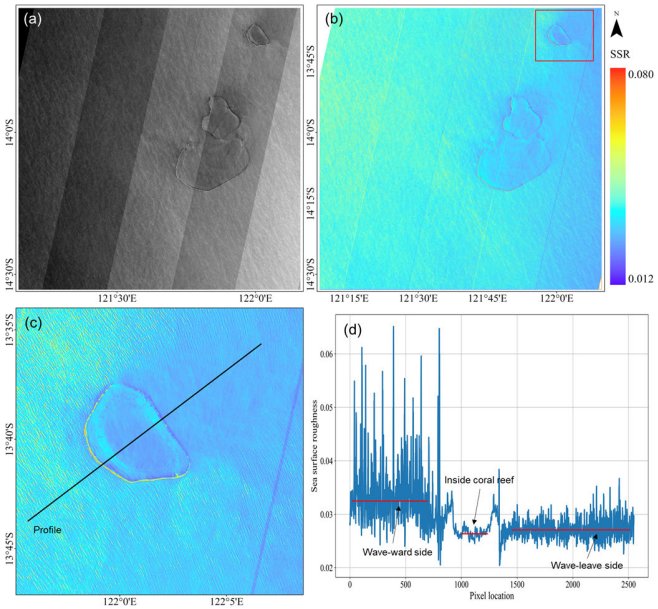


Fig. 11. (a) ρ_g image in the NIR band. The satellite image acquired by Sentinel-2A on October 9, 2020, located in the Timor Sea. (b) SSR-estimated image of coral reefs. (c) Magnified SSR image of the red box in (b). (d) SSR of the "Profile" line in (c). The red line indicates the mean SSR of the wave-ward side of the reef (0.033), inside the coral reef (0.026), and the wave-leaving side of the reef (0.027).

inner reef reduces SSR by 0.007 or 21% of the background SSR generated by wind. The SSR in the direction in which the waves leave gradually increases. These results reveal that coral reefs both reduce SSR and determine wave direction by altering the trend in SSR.

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2) English Channel: In Fig. 12(a), we present an SSR image acquired by Sentinel-2A on June 26, 2018, over the English Channel, in which several ship wakes and the Rampion offshore wind farm are visible. As seen in the figure, the longest ship wake is >20 km. Fig. 12(b) shows a magnified SSR image of the red box labeled S1 in Fig. 12(a). The ship wake changes the sea surface wave slope, which affects the SG. A complete Kelvin wake consists of transverse, divergent, and cusp-shaped waves formed by the coherence of the transverse and divergent waves [37]. Because of the short wavelength of cusp waves, they are often shaped such as V-shaped lines in satellite images, also known as Kelvin's arms [38]. In Fig. 12(b), we can see the transverse waves and Kelvin arms, as well as the turbulent wake. The SSR of the main wake is very low compared to its surroundings, which is due to the fact that the ship has changed the original sea surface imaging geometry, and the SG intensity is suppressed. The change of SSR suggests that quantitative detection of Kelvin wakes can be achieved, with the potential to recognize ships, which is of great significance in the field of fisheries management and maritime surveillance [39], [40].

Fig. 12(c) shows a magnified SSR image of the red box labeled S2 in Fig. 12(a), which is the Rampion offshore wind farm, approximately 15-km offshore from Sussex, U.K., with 116 wind turbines. The black arrow indicates wind direction; the mean wind speed was 3.95 m/s. SSR shows significant spatial differences. This illustrates the complexity of offshore sea surface conditions, which are influenced by phenomena such as tides, currents, and human activities. Vanhellemont and Ruddick [41] demonstrated turbid wakes consistent with the direction of the tidal current using Landsat-8 images. As shown in Fig. 12(c), we can observe two types of wake on the turbine side: tidal-current wake and wind-driven turbulence wake. The length of the tidal-current wake is longer, ranging from one to several kilometers, with a maximum length of more than 7 km, which is probably because the wakes continue to extend after the tidal-current changes direction. The direction of the turbulence wake is consistent with the wind direction. This may be a current field Kármán vortex street caused by a stabilizing wind field. Based on the variation in the SSR, quantitative detection of the flow and wind fields around the wind farm can be realized, which is of great significance for the monitoring of wind farms [42], [43]. Based on the above results, the advantages of high-resolution SSR images (30 or 10 m) for marine applications are evident, enabling the observation and analysis of small-scale sea surface features.

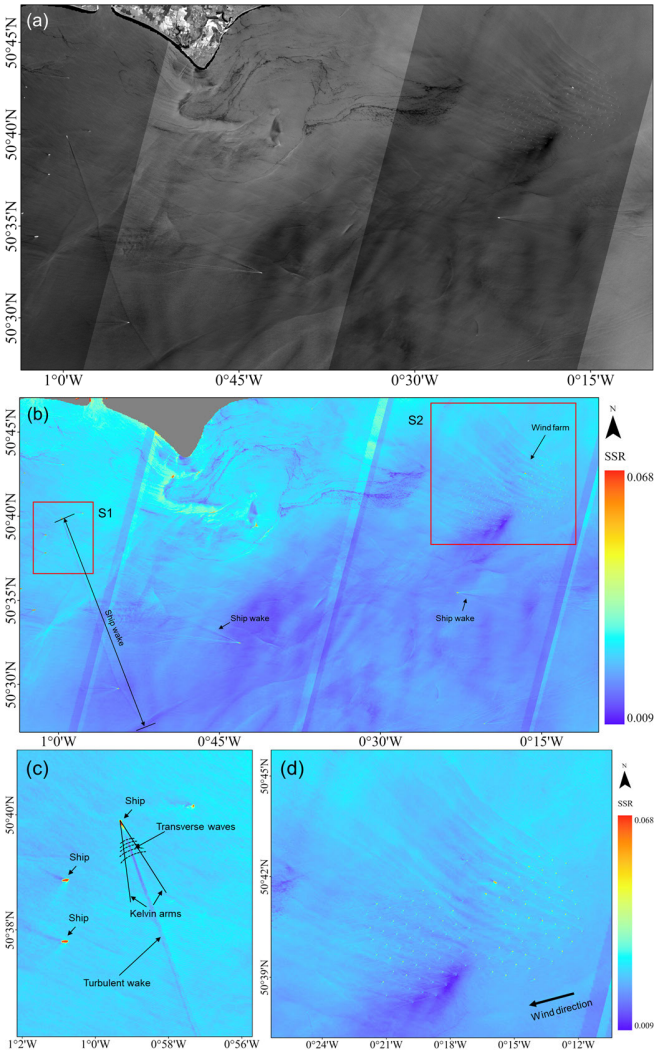


Fig. 12. (a) ρ_g image in the NIR band. The satellite image acquired by Sentinel-2A on June 26, 2018, over the English Channel. (b) Estimated SSR image showing ship wakes and offshore wind farm. (c) Magnified SSR image of the red box labeled S1 in (b), the Kelvin wake generated by the moving ship can be seen. (d) Magnified SSR image of the red box labeled S2 in (b), which shows the Rampion offshore wind farm, approximately 15 km offshore from Sussex, U.K. The black arrow indicates the wind direction, and the mean wind speed is 3.95 m/s.

V. Discussion

A. Sensitivity of Seawater Refractive Index

In Section III-B, we choose the refractive index as the output parameter for SSR inversion. The refractive index of seawater in the open ocean is relatively fixed at specific wavelengths; however, the seawater temperature and salinity affect the seawater refractive index [44]. Röttgers et al. [29] integrated the variation in the refractive index with temperature and salinity in the range of 300–4000 nm based on the studies by Quan and Fry [45], and Max and Chapados [46]. Therefore, a sensitivity analysis of the refractive index is performed. The refractive index of the NIR band (∼865 nm) ranges from 1.327 to 1.334, and that of the SWIR band (∼1610 nm) ranges from 1.318 to 1.325 when the salinity varies from 0 to 30 psu and the temperature varies from 0° to 30°. The effect of the refractive index is analyzed by comparing the wind speed converted by the SSR with the reanalysis wind speed; the results are listed in Tables III and IV.

As can be seen from Tables III and IV, when the refractive index increases in both bands, the MAE and RMSE increase slightly, and R remains essentially unchanged. Overall, the results show that a change in the refractive index has a smaller effect on the SSR results.

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TABLE III
REFRACTIVE INDEX SENSITIVITY ANALYSIS IN THE LANDSAT-8 NIR BAND

	1.327	1.33 (this study)	1.334
R	0.89	0.88	0.87
MAE (m/s)	0.62	0.64	0.67
RMSE (m/s)	0.80	0.83	0.88

TABLE IV
REFRACTIVE INDEX SENSITIVITY ANALYSIS IN THE LANDSAT-8 SWIR BAND

	1.318	1.32 (this study)	1.325
R	0.89	0.89	0.88
MAE (m/s)	0.59	0.59	0.63
RMSE (m/s)	0.75	0.76	0.81

B. Uncertainty of SSR Ambiguity Elimination Method

In Section III-C, we describe a simulation experiment of ambiguity, demonstrated that Landsat-8 and Sentinel-2 images can cause ambiguity in SSR retrieval, and explored the causes of SSR ambiguity and its distribution characteristics. Therefore, we propose an ambiguity critical point identification and ambiguity elimination method for SSR estimation based on single-angle SG images that realizes a pixel-scale quantitative inversion of SSR. The key feature of this method is that the median roughness in the unambiguous region is the input for the SG simulation. However, the actual sea surface is complex and is affected by various random fluctuation signals. The spatial distribution of SSR in the satellite images is inconsistent. Thus, the median roughness does not accurately replace SSR at the critical position and may introduce uncertainty errors.

In Section IV-A, we present a case of ambiguity (Fig. 6). The point "p5" in Fig. 6(d) is the critical position extracted by this method, but it can be found that SSR1 and SSR2 of the pixels near "p5" are almost overlapped, which is different from the single point of overlap between SSR1 and SSR2 of "p5" in Fig. 5(e). Therefore, assuming that "p5" in Fig. 6(d) is shifted by 100 pixels to the left or right, it is difficult to quantify whether or not the results will be enhanced (reduced). Summarizing the results of this study, the method may not be optimal; however, it performed well for SSR estimation and wind-speed validation.

C. Influence of Observation Geometry

Fig. 3 illustrates a case where ambiguity occurs when retrieving SSR at a Sun-surface-observer observation geometry using SG. Based on the analysis of Fig. 3(b), it can be seen that whether the ambiguity phenomenon occurs or not is affected by the SG intensity in addition to the observation geometry. Therefore, it is difficult to obtain a generalized observation geometric relationship for describing ambiguity based on Fig. 4. However, we can describe the observation geometries (which may lead to ambiguity) for specific satellites.

Fox et al. [17] used the minimum Sun glitter angle (SGA) of MISR to describe the observation geometrical relationships that lead to possible ambiguities. Statistically, when the minimum SGA is less than 6°, SSR is determined unambiguously; when it is greater than 6° or 7°, the model slopes are not monotonic and may produce ambiguous solutions [17]. Jackson and Alpers [18] further investigated the relationship between Sun glitter and sea surface tilt angle (β) on the basis of Melsheimer and Kwoh [47]. According to Fig. 3(b), it can be seen that for a particular viewing geometry, the Sun glitter is not a simple monotonic function of the sea surface roughness but has a maximum at $\sigma_0^2 = \tan^2 \beta$ when the wave-slope probability density function is symmetric [18].

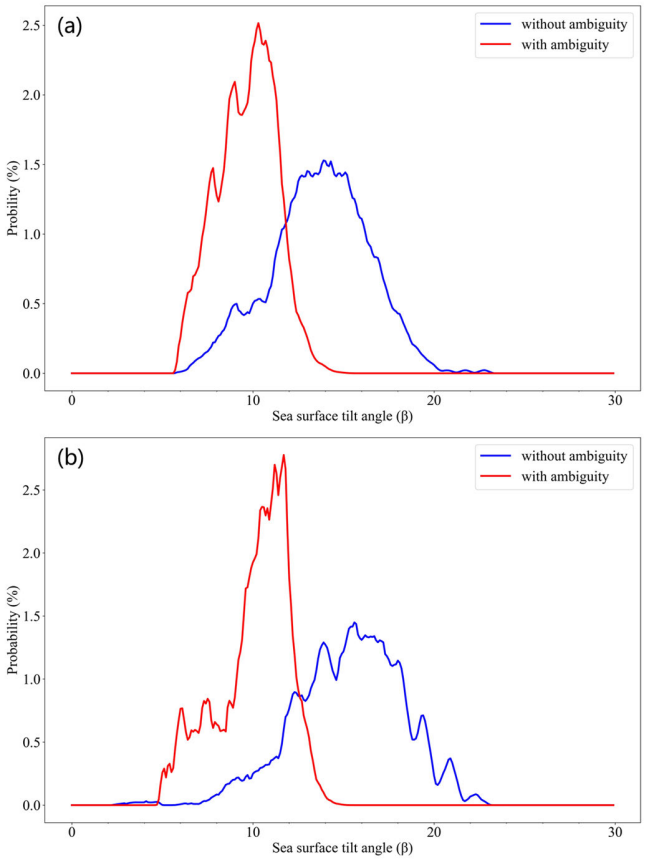


Fig. 13. Probability distribution of β where with and without ambiguity pixels in (a) Landsat-8 and (b) Sentinel-2 images.

As described in Section III-A, β is affected by θ_0, θ_v, and $\Delta\emptyset$. Therefore, based on the studies of Fox et al. [17] and Jackson and Alpers [18], we can utilize β to describe the observation geometrical relationships that lead to possible ambiguities for Landsat-8 and Sentinel-2. We count β where ambiguity and no ambiguity pixels of Landsat-8 and Sentinel-2 images, respectively. Then, we establish histograms of the occurrence of ambiguity and no ambiguity as a function of β for both Landsat-8 and Sentinel-2 observations. The probability distribution of β is shown in Fig. 13. According to Fig. 13(a) and (b), β of Landsat-8 images ranges from 6° to 20°, and that of the Sentinel-2 images ranges from 4° to 24°. For Landsat-8 images, when β is greater than 14°, no ambiguity occurs; at 6° to 14°, ambiguity can occur. For Sentinel-2 images, there is no ambiguity when β is greater than 14°; between 4° and 14°, ambiguity occurs.

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VI. Conclusion

In this study, a method for adaptive estimation of SSR at the pixel scale for single-angle SG images is developed, and the results are summarized as follows.

1) We explore the causes of SSR ambiguity and its distribution characteristics using simulations. Based on this, we propose an ambiguity critical point identification and ambiguity elimination method for SSR estimation based on single-angle SG images, which realizes the pixel-scale adaptive quantitative inversion of SSR. We selected more than 200 SG images from Landsat-8 OLI and Sentinel-2 MSI for SSR estimation and analyzed SSR cases.
2) Based on the linear relationship between SSR and wind speed, wind speed is chosen as the metric for verifying the model's accuracy. Estimated, reanalysis, and buoy wind-speed data (the latter from ECMWF and NDBC, respectively) are altitude-adjusted and time- and position-matched. Comparing the estimated and reanalysis wind speeds produces an RMSE of ca. 1.0 m/s and MAE <1.0 m/s, and estimated and buoy wind speeds with an RMSE of 2.24 m/s and MAE of 1.79 m/s. Based on these results, the method proposed here can effectively eliminate the ambiguity in SSR and improve the accuracy of high-resolution SSR data.
3) High-resolution SSR data have significant advantages in the quantitative analysis of sea surface dynamics features. The method proposed here not only removes the ambiguity in SSR but also expands the application prospects for publicly available global high-resolution optical satellite imagery. To illustrate its application potential, we exhibit and analyze typical oceanographic cases for two regions, the Timor Sea and the English Channel. These findings reveal the importance of SSR data for monitoring marine phenomena (internal waves, coral reefs, ship wake, offshore wind farms, and so on), as well as its potential application in further quantifying the depth of internal waves, shallow-sea topography detection, and so on.

In summary, the method proposed in this study has high applicability for SSR ambiguity eliminating different satellites (Landsat-8 OLI and Sentinel-2 MSI) or bands (NIR and SWIR) without relying on other auxiliary parameters. In particular, the proposed method furthers the use of single-angle SG images to retrieve SSR at the pixel scale. In addition, the accurate estimation of high-resolution SSR results has immeasurable potential for monitoring ocean internal waves, coral reef barrier waves, ship wakes, and offshore wind farms. However, this method relies on high-precision Sun glitter signals. Some factors (e.g., nearshore turbid water and thin clouds) can affect the extraction of Sun glitter. If the satellite does not have an NIR band or short-wave infrared band, the Sun glitter signal from nearshore turbid water may have a large error. Also, this method does not work if the sea surface is covered by a nonwater body (e.g., oil film). Because the refractive index of the oil film is unknown, the inversion of SSR cannot be realized. In the future, we plan to analyze the response of high-resolution SSR to different ocean phenomena such as ocean fronts, river plumes, and ocean eddies.

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